

Introduction to the Data Analytics

Statistical Toolbox

Summary

- **Concepts (25 mins)**
 - Where Most Data Analysis Goes Wrong
- **Tools (25 mins)**
 - Purpose
 - Examples
- **Questions (10 mins)**

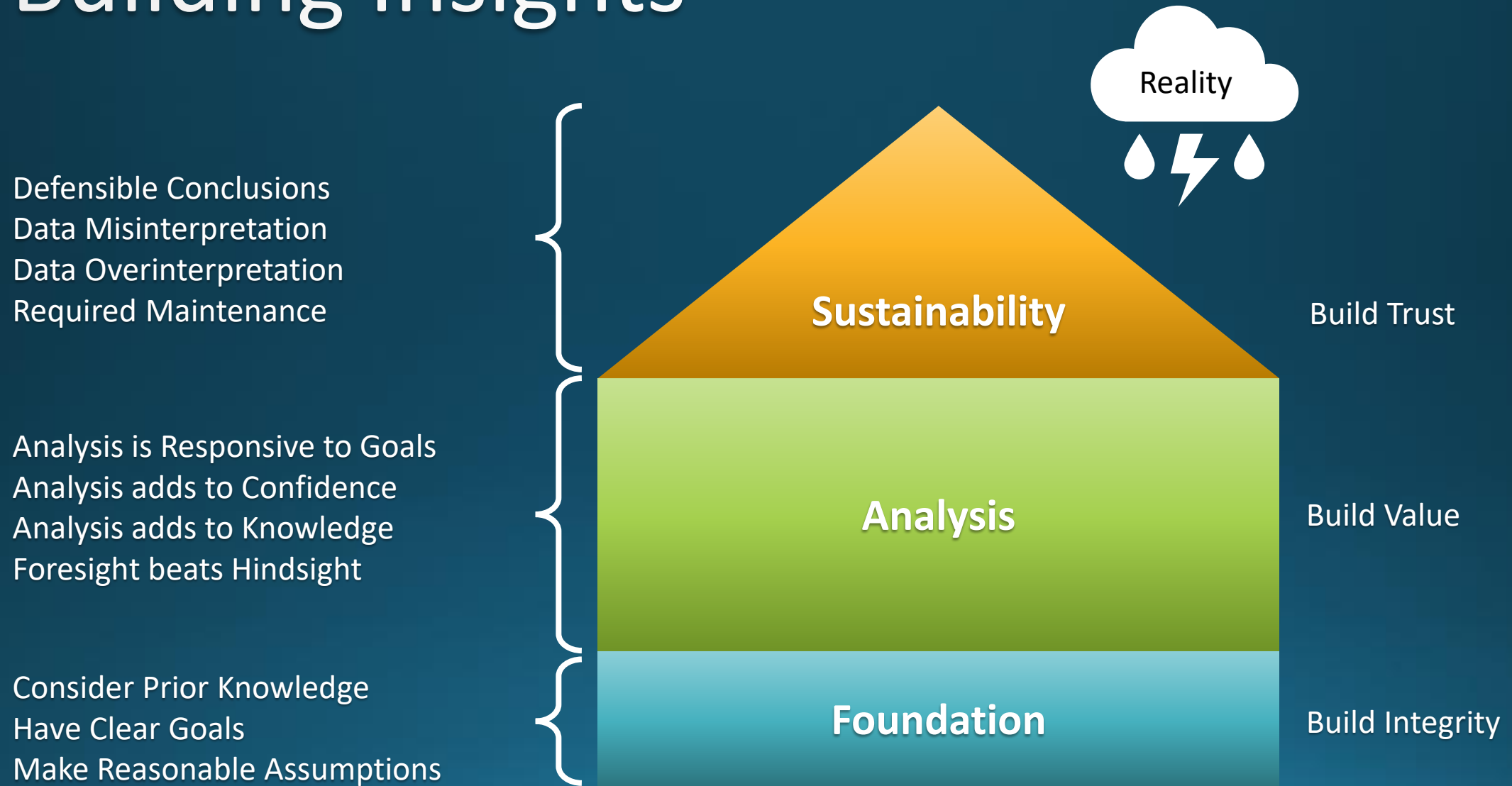


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Foundational Statistics

Concepts

Building Insights



Statistics as a Toolbox

- **What is Statistics?**
 - A scientific toolset for collecting, analyzing and interpreting data.
- **Why is it Important?**
 - Used to make inferences and understand uncertainty.

Numbers

- **Continuous**

- Unlimited Possible Values/Outcomes
 - What is the temperature outside? 74.4°F, 86.32539948°C
 - How tall are you? 71.4 in, 182.883 cm

- **Discrete**

- Limited Possible Values/Outcomes
 - How many children do you have? 0, 1, 2, 3
 - What size shoe do you wear? 8, 8.5, 9, 9.5

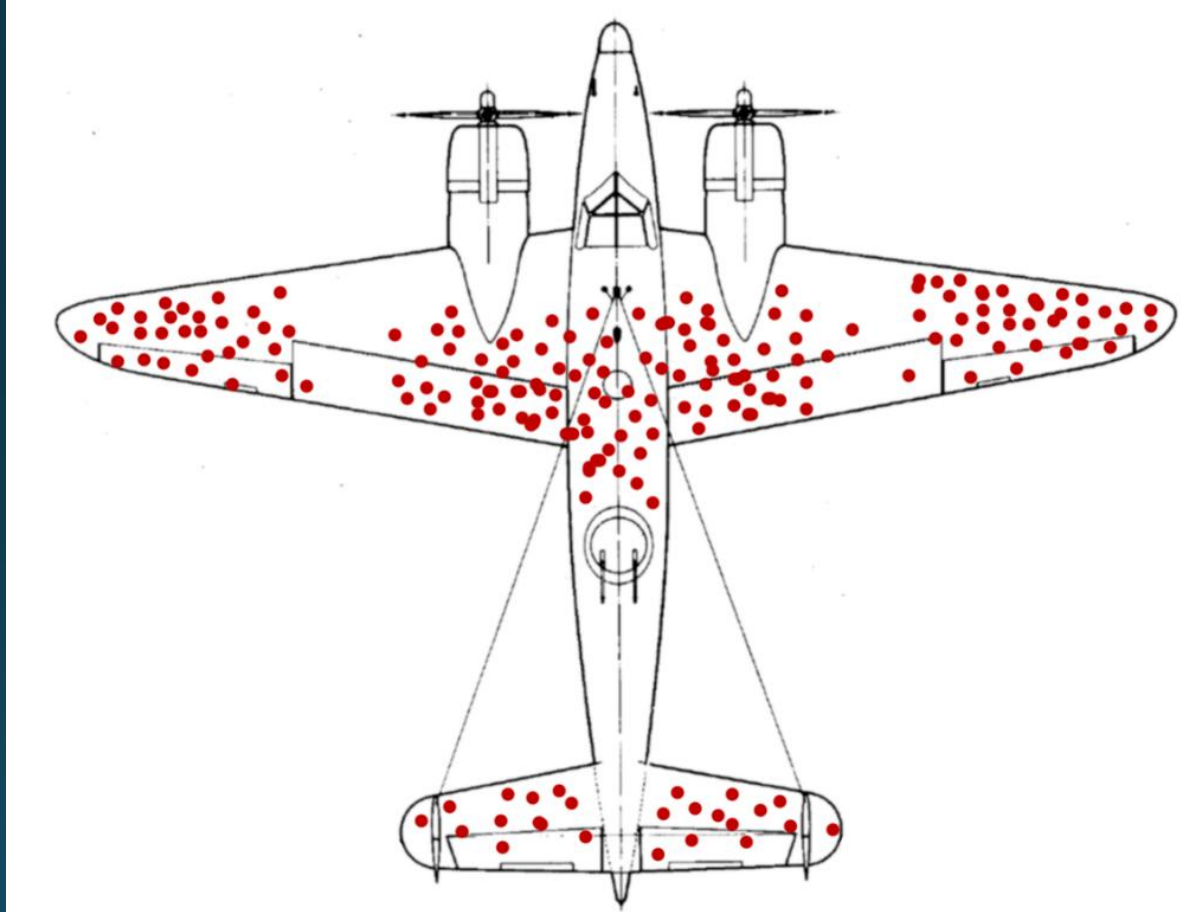
Data vs. Process

- What are **Data**?
 - Collection of observations (not facts).
 - Data are subject to imperfect observers and instrumentation.
 - Outcomes of one or more processes.
- What is a **Process**?
 - Actions, phenomena, and/or mechanisms that lead to creation, collection, or generation of data.
- Data is inherently tied to processes that generated it.
- Data reflects (or is result of) underlying processes.

Sample vs. Population

- What is a **Sample**?
 - I have SOME observations.
 - Always shoot for 30+ observations. The more the better.
 - In most cases, we work with samples.
- What is a **Population**?
 - I have ALL the observations.
 - In rare cases, we work with populations.

Summarize With An Example



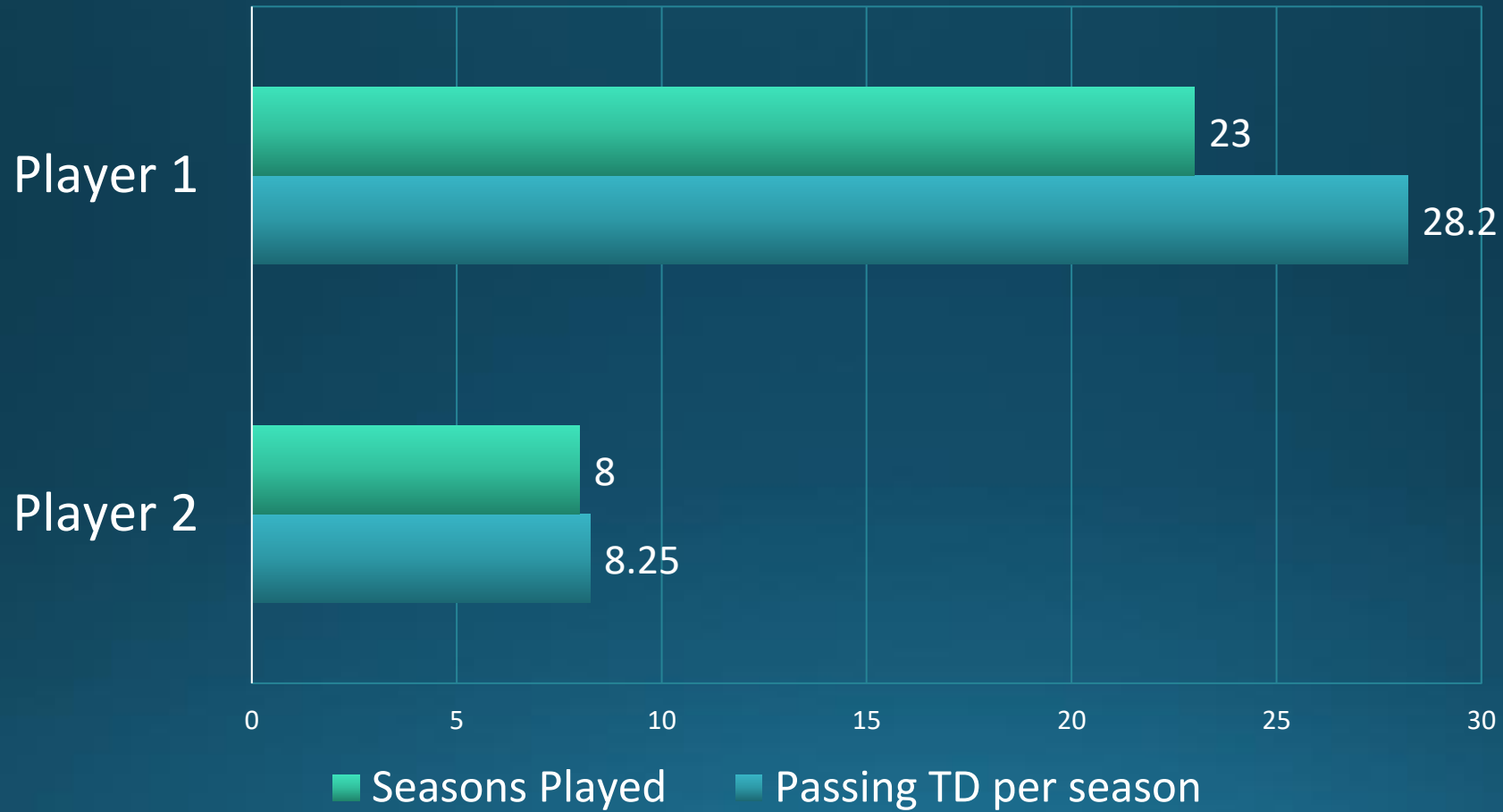
- **Sample:** some observations
- **Population:** the whole picture
- **Data:** measured outcomes
- **Process:** creates the data

Process Stationarity

Does your process change over time?

- ✓ Required assumption for most statistical tools
 - ✓ Property of the process generating the data
 - ✓ Depends on how data are grouped.
 - ✓ Exists on a spectrum (Strong to Weak to Non-stationary).
-
- Not a characteristic of the data.
 - There is no simple/definitive test for process stationarity.

Who is the Better Quarterback?

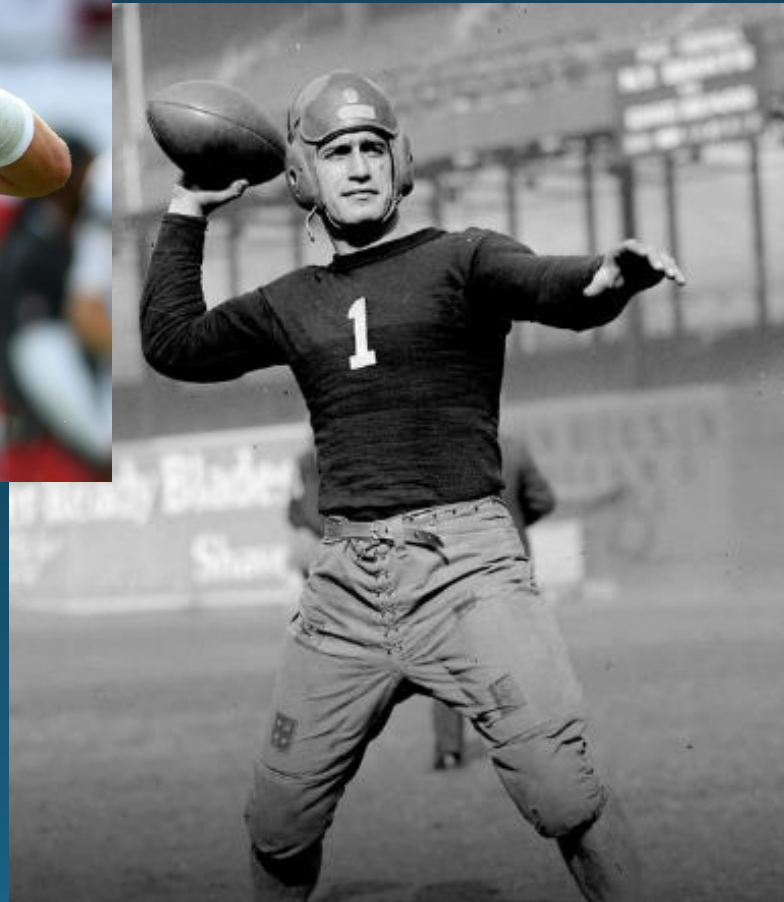


Consider the Process



Player 1: Tom Brady (2000-2022)

Player 2: Benny Friedman (1927-1934)



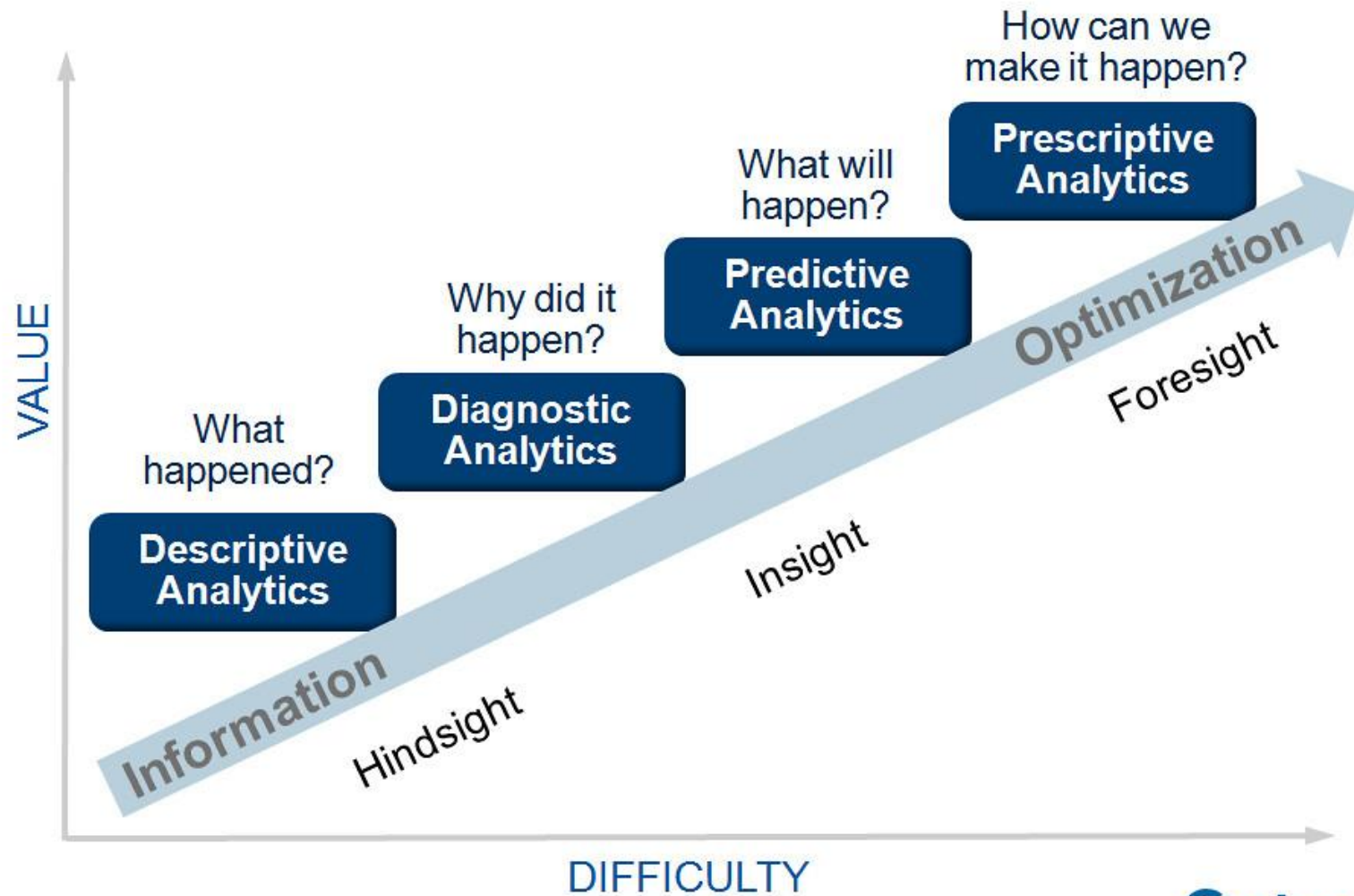
- Process Factors
 - Length of Season
 - Rules / Penalties
 - Pads / Helmet
 - Offensive Line

Do the collective processes seem stationary over time?

Foundational Statistics

Tools

Analytic Value Escalator



Examples

Descriptive	Diagnostic	Predictive	Prescriptive
Distributions	Correlation Analysis	Linear Regression	Optimization Algorithms
Histograms	Regression Analysis	Logistic Regression	Decision Analysis
Central Tendency Metrics	Chi-Square Test	Time Series Analysis	Simulation
Dispersion Metrics	Variance Analysis	Decision Trees	Markov Chains
Box Plots	Normality Tests	Random Forest	Game Theory
Z Score	Residual Analysis	Neural Networks	Genetic Algorithms
Standard Error	Homogeneity Tests	Geostatistics (e.g., Kriging)	Reinforcement Learning
		Bayes Theorem	Bayes Theorem

Distributions

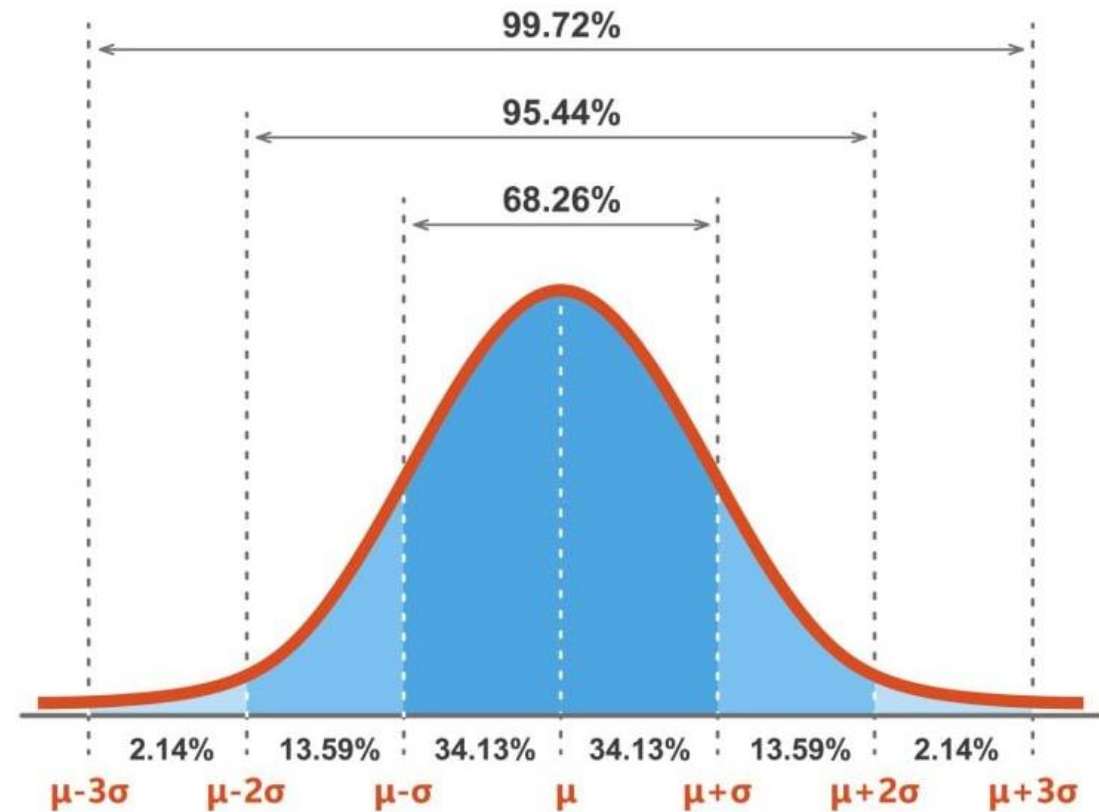
Goal:

- Describe the **frequency** of values for one **continuous variable** within a data set.

μ = Mean

σ = Standard Deviation

Height of Male Firefighters- a Normal Distribution

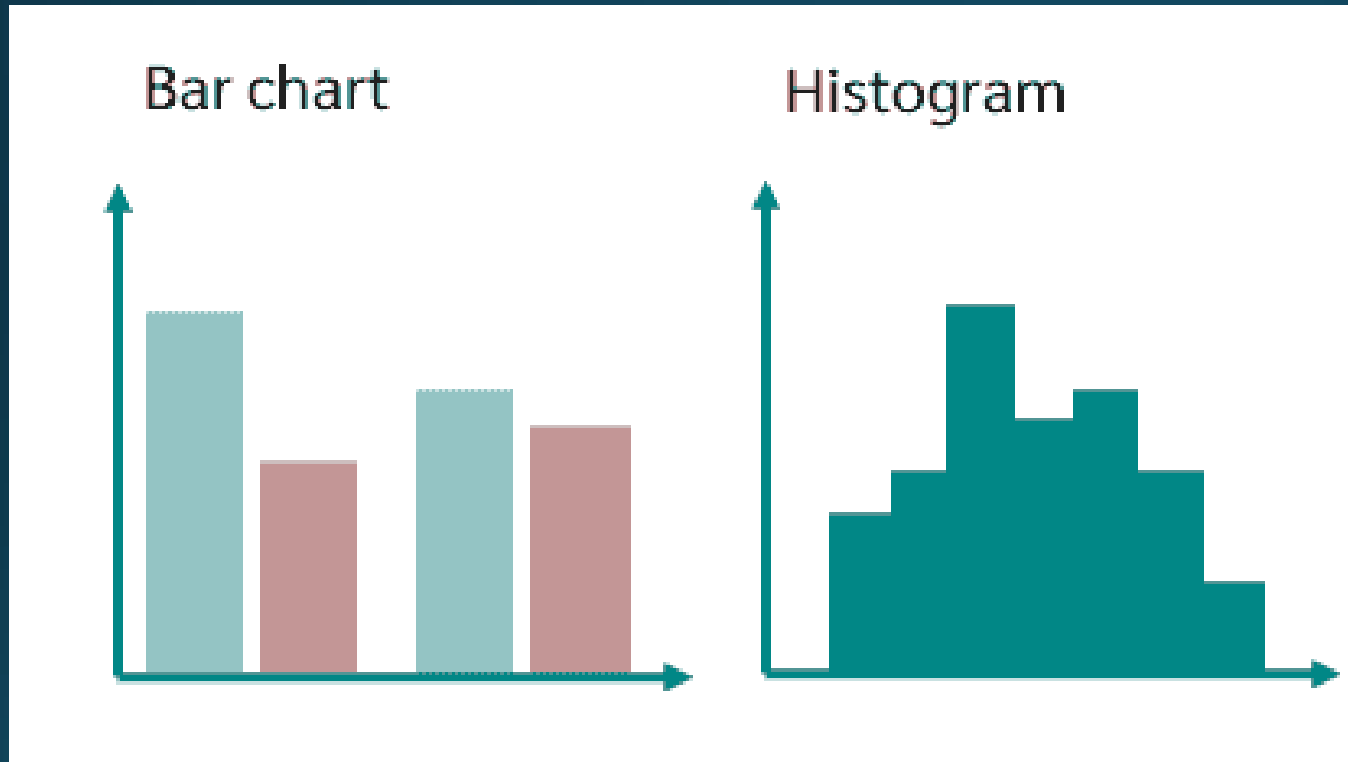


5'1"
= $\mu - 3\sigma$

5'9" = Average
= μ

6'5"
= $\mu + 3\sigma$

Bar Charts vs. Histograms



Discrete Categories

Proportional Bar Height

Continuous Distributions

Proportional Bin Area

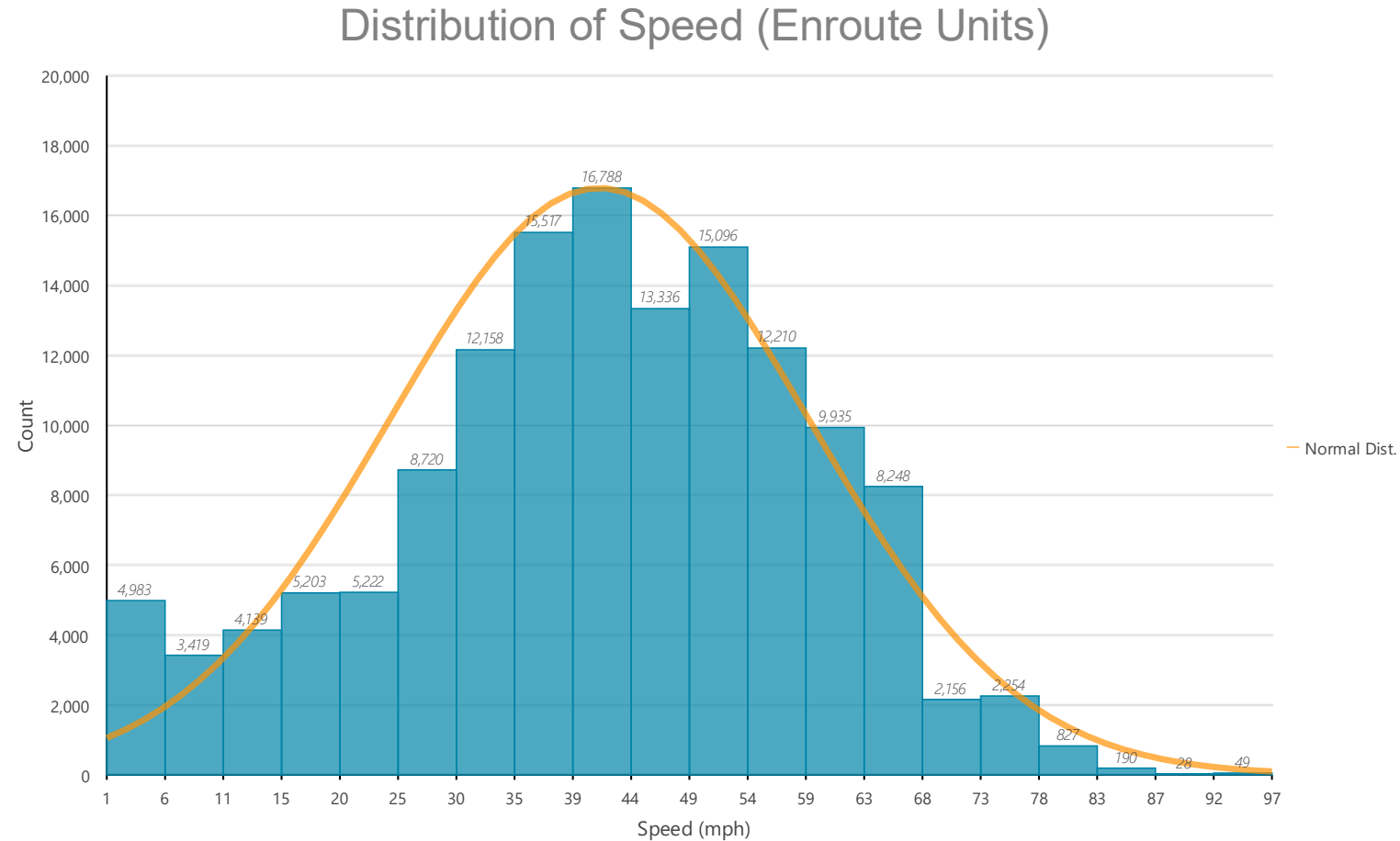


Can do just about anything.

Example

Describe frequency of vehicle speed while engines are enroute to incidents.

2-weeks of data (April 2024).



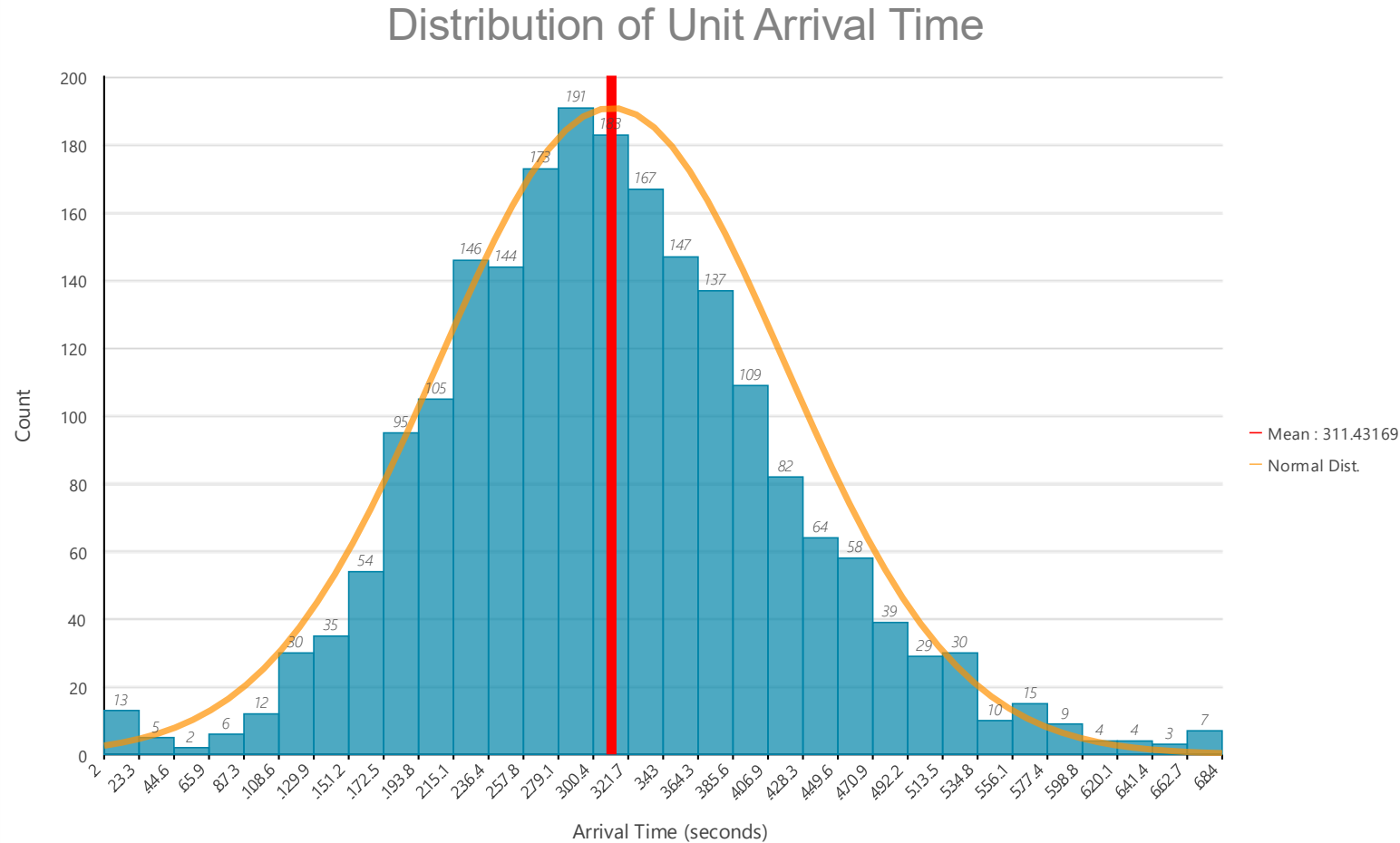
Central Tendency

- Mean, Median, Mode
- **Continuous Normal Distribution:** Mean
- **Continuous Skewed Distribution:** Median
- **Discrete Distribution:** Mode

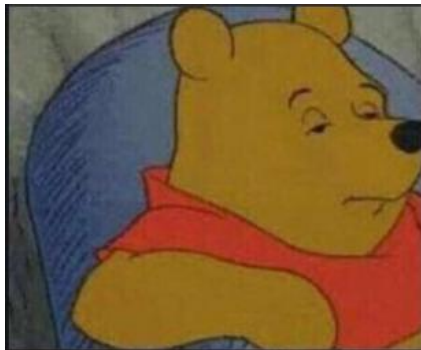
Example

Describe frequency of unit arrival time to incidents.

Red bar = Mean



Other Distribution Types



BELL CURVE



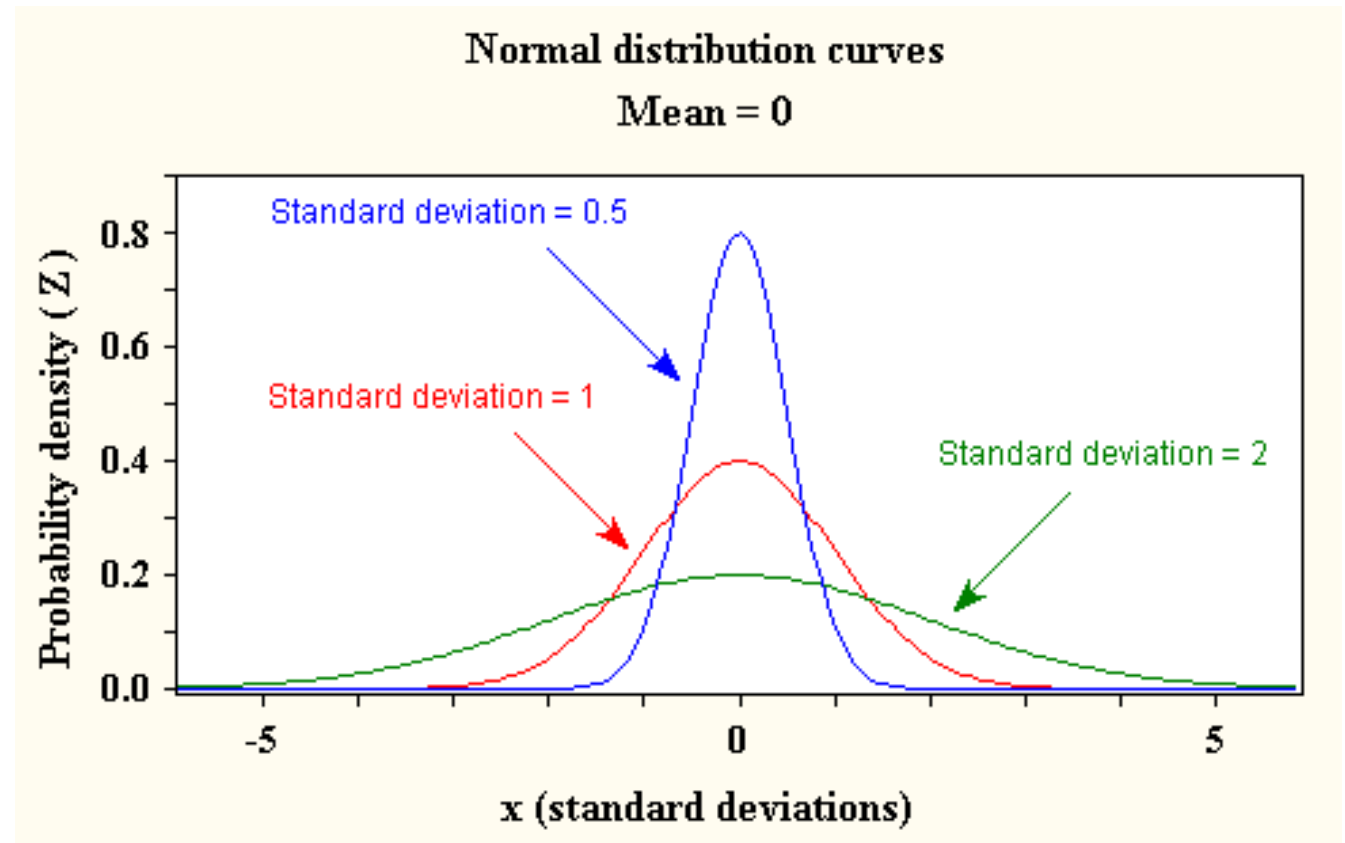
**GAUSSIAN
DISTRIBUTION**

- **Process Models**

- Normal (Gaussian)
- Chi-square
- Binomial
- Poisson
- Uniform
- Log-normal

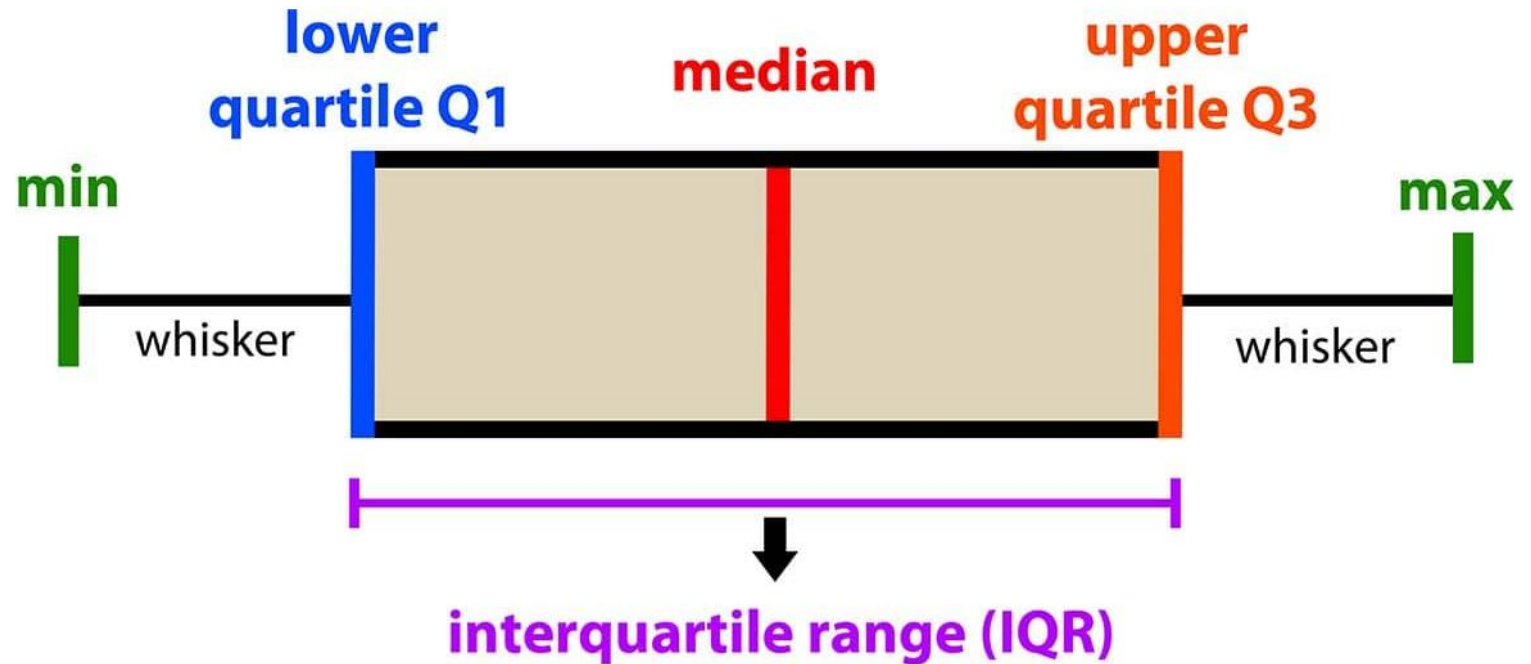
Measures of Dispersion

- Range
 - Minimum Value
 - Maximum Value
- Standard Deviation
- Variance



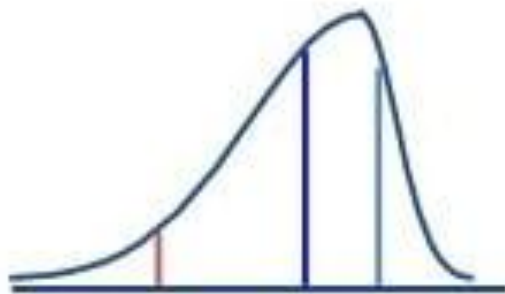
Box Plots

- Useful for quick description about a distribution.
- Useful for comparing different distributions.

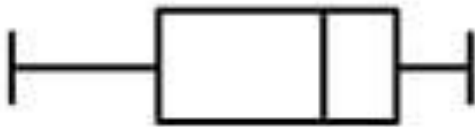


Distribution Skew

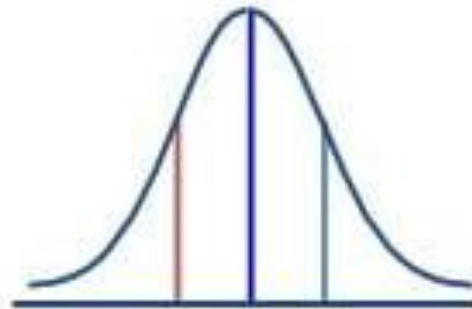
Left-Skewed



Q_1 Q_2 Q_3



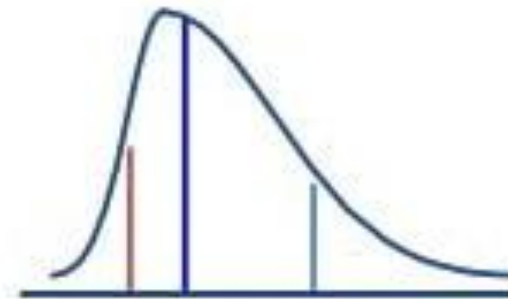
Symmetric



Q_1 Q_2 Q_3



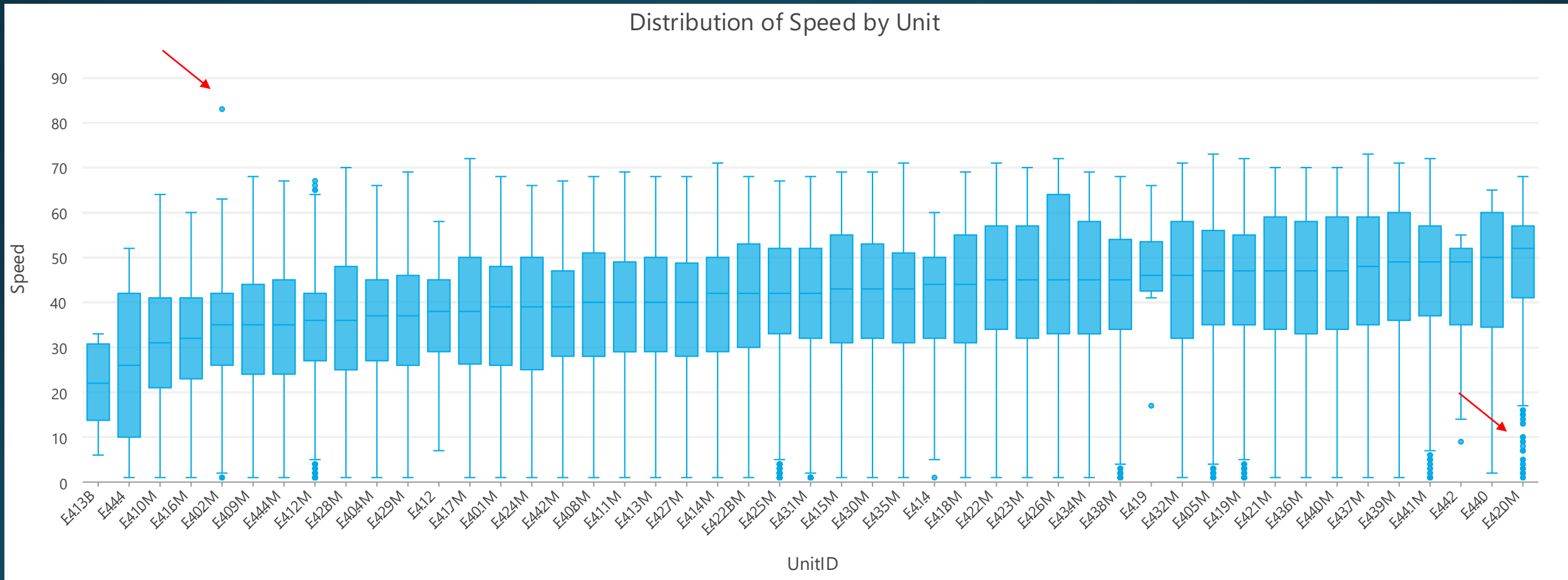
Right-Skewed

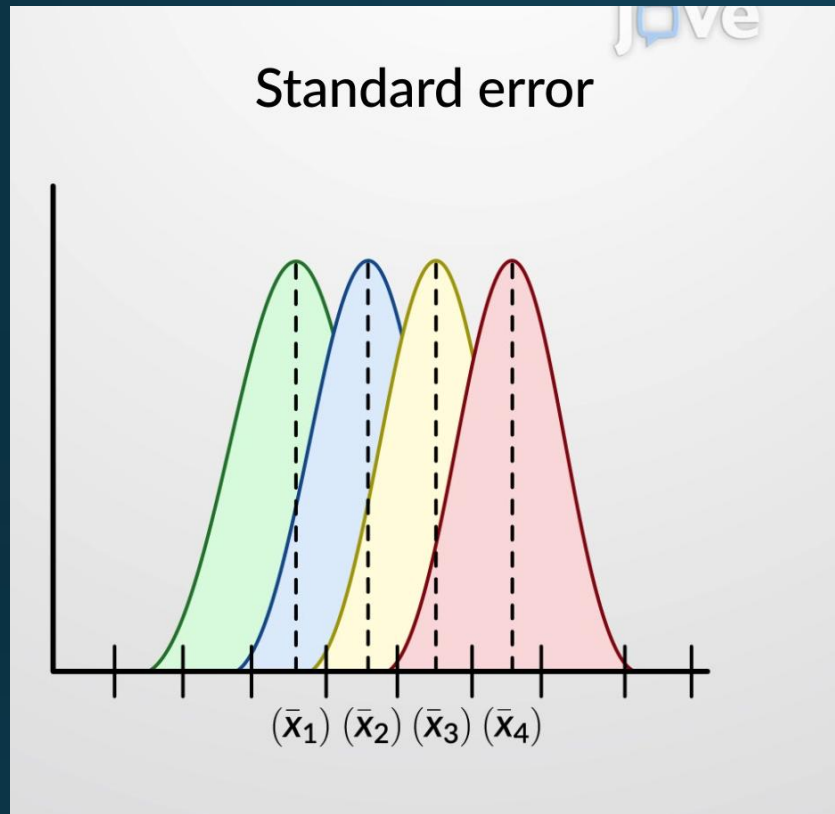


Q_1 Q_2 Q_3



Consider the Process





$$SE = \frac{\sigma}{\sqrt{n}}$$

← Standard deviation

← Number of samples

“to what degree
do the sample
means vary”

Standard Error

- Quantifies how much **sample mean is likely to vary from the true population mean** if multiple samples were taken from same population.
- Measure of Data's Uncertainty

Confidence Interval

- Turns uncertainty into a range
- Wide CI → low precision
- Narrow CI → high precision
- Z is commonly 95%

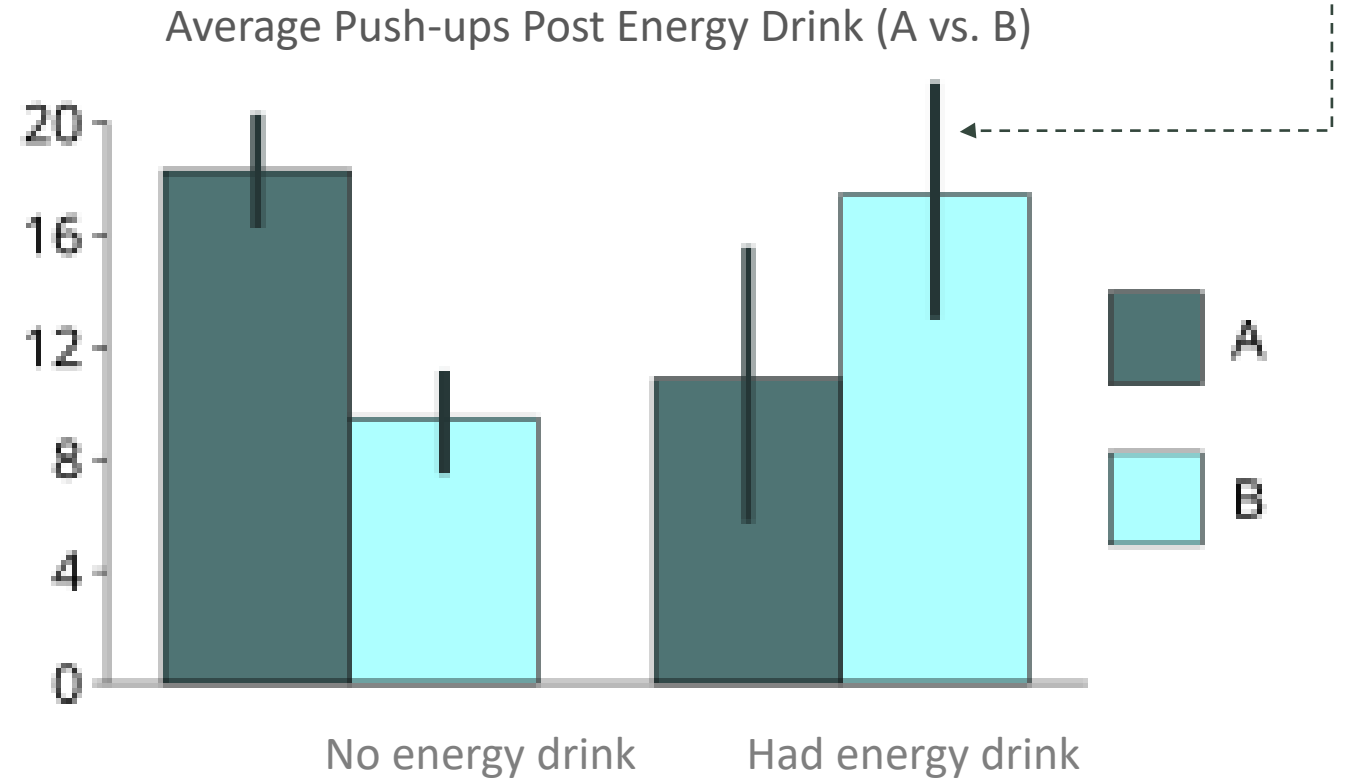
$$CI = \bar{x} \pm (Z * SE)$$

CI = confidence interval

$\bar{x}$ = sample mean

Z = confidence level value

SE = standard error of the sample mean



Communicating Results

- “All models are wrong, but some are useful”



Thank you!



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